

International Energy Agency (IEA) Global Energy Efficiency and Transition Report

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Chapter 1: Global Energy Intensity Trends and Net Zero Milestones

Global primary energy intensity improved by 2% in the preceding evaluation year, falling short of global climate targets.

To remain on track for the Net Zero Emissions by 2050 scenario, annual energy intensity improvement must accelerate to 4% p

Energy efficiency represents the first fuel of the clean transition, delivering immediate emissions reductions and energy security

Policy adoption of mandatory building energy codes and appliance minimum performance standards covers 45% of global build

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Chapter 2: Electrification and Heat Pump Technology Adoption

Global sales of residential and commercial heat pumps expanded by 11%, driven by European and North American policy incentives.

Heat pumps provide thermal efficiency up to 400% higher than conventional fossil gas boilers by extracting ambient thermal energy.

Electric vehicle (EV) sales accounted for 18% of all new cars sold globally, reducing crude oil consumption in passenger transport.

Grid infrastructure modernization and smart metering deployment are vital to balance bidirectional renewable electric loads.

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Chapter 3: Industrial Thermal Efficiency and Waste Heat Recovery

Industrial waste heat recovery systems in steel, cement, and chemical plants can reduce industrial fuel consumption by up to 15%.

Implementing high-efficiency electric motors and variable-frequency drives saves industries up to 300 terawatt-hours of power annually.

Governments should establish fiscal tax credits for capital investments in industrial heat exchangers and thermal storage.

Digital energy management systems (EMS) utilizing IoT sensors optimize factory load balancing and shave peak electricity demand.